

Docling

The Open-Source Engine for AI-Ready Documents

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Advisory Software Engineer @ IBM



Agenda

1. Motivation
2. Quick tour
3. Deep dive
4. Roadmap
5. Wrap-up

Motivation

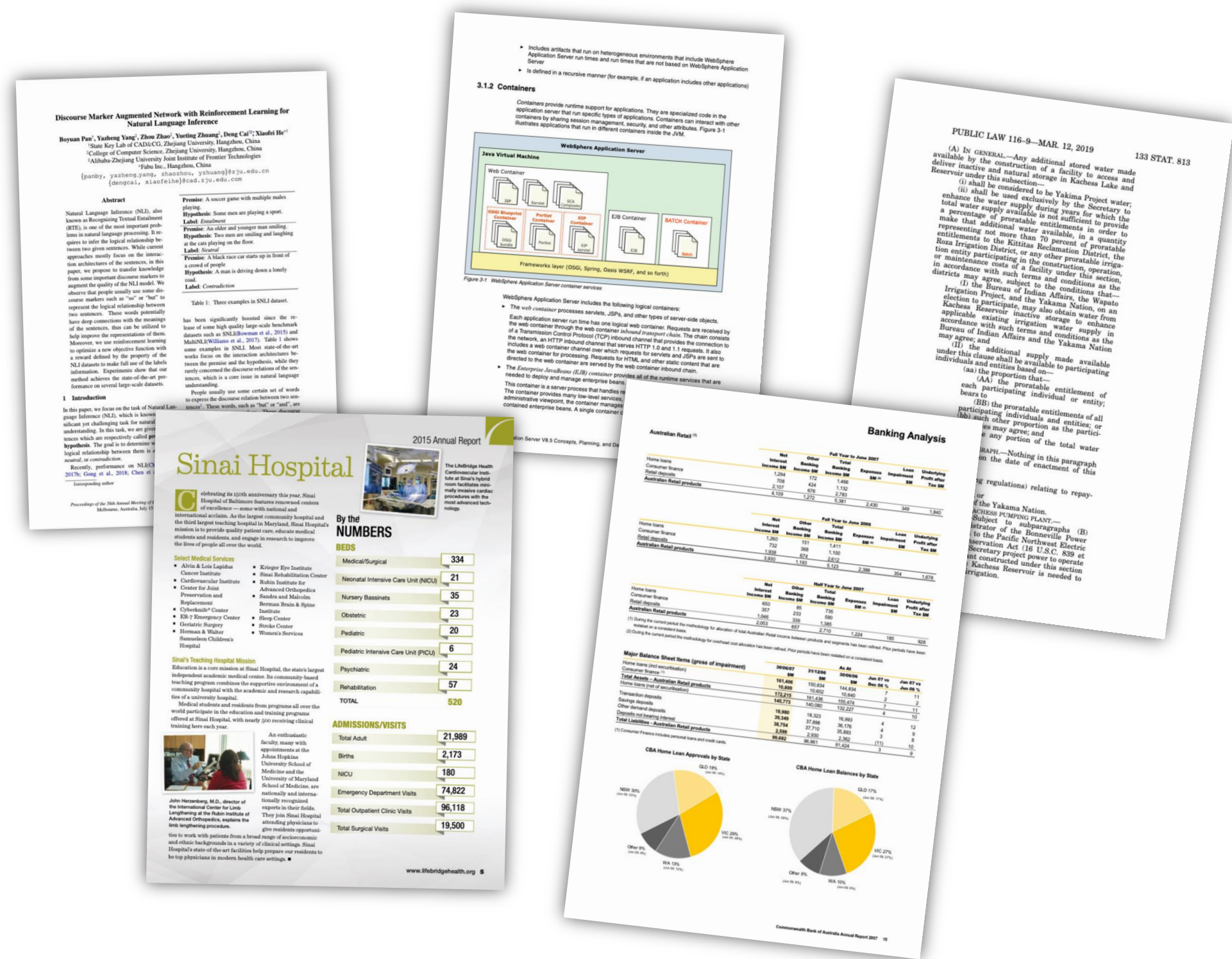
Documents & challenges

- Documents are everywhere & come in various flavors & formats

- Working with documents is complicated

- LLMs do not understand documents natively

- Bridging the gap typically implies significant sacrifices



Meet Docling



Open-source
& MIT-licensed



Advanced, cost-efficient
document intelligence



Multi-format
support



Unified, rich, LLM-friendly
representation



Privacy & security
via local execution



Plug-and-play integration
for building apps & agents

Quick tour

Birgit Pfiftzmann, Christoph Auer, Michele Dolfi, Ahmed S. Nassar, and Peter Staar. 2022. DocLayNet: A Large Human-Annotated Dataset for Document-Lay-out Analysis. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)*, August 14–18, 2022, Washington, DC, USA. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3534678.3539043>

Core team & contributors



docling

Feb 23, 2025 → Feb 23, 2026

Active contributors

2,159 📈 192.2% (+1,420)
vs. 739 last period

🛡️ Maintainers 👁️ Reviewers
15 59

weekly active contributors



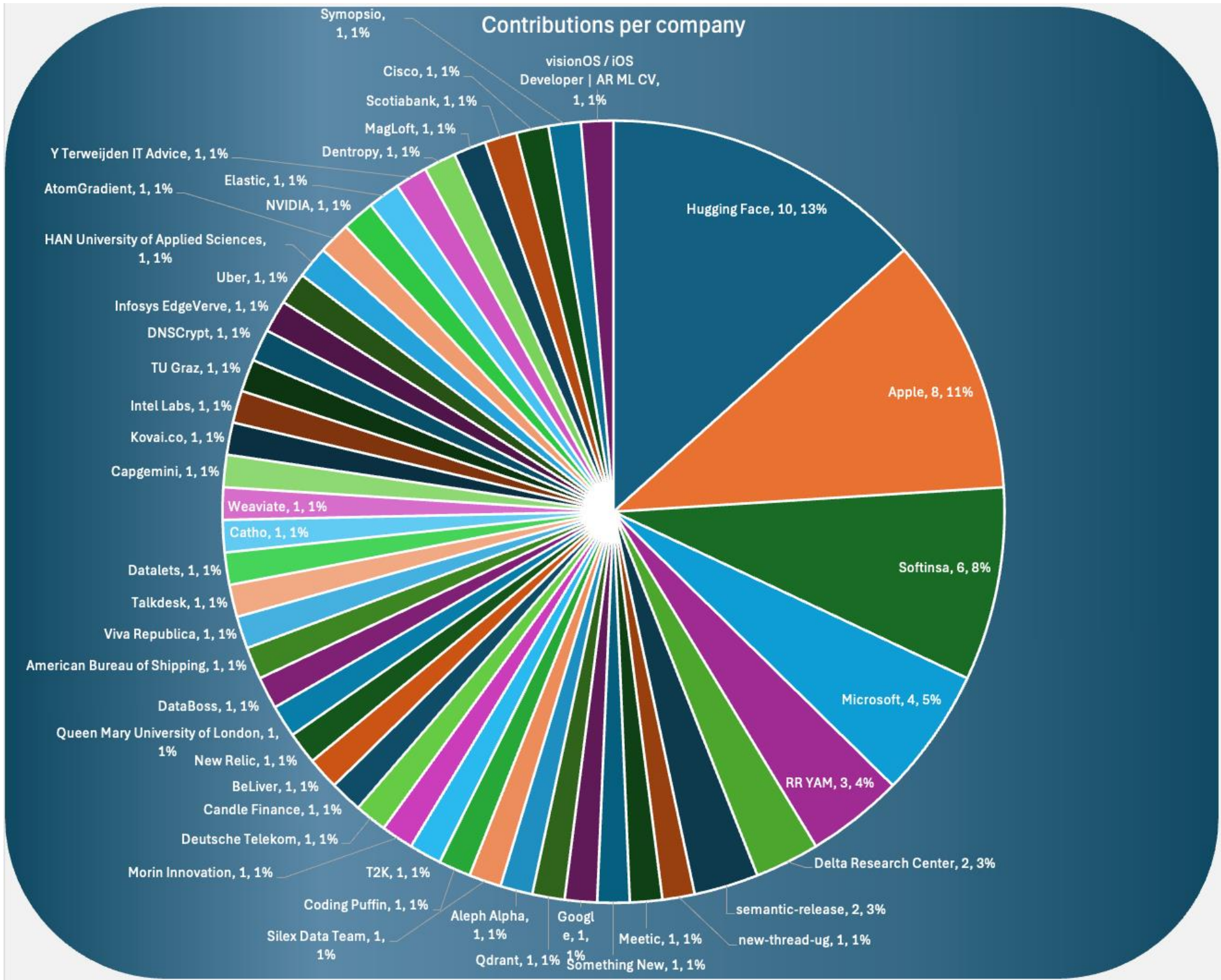
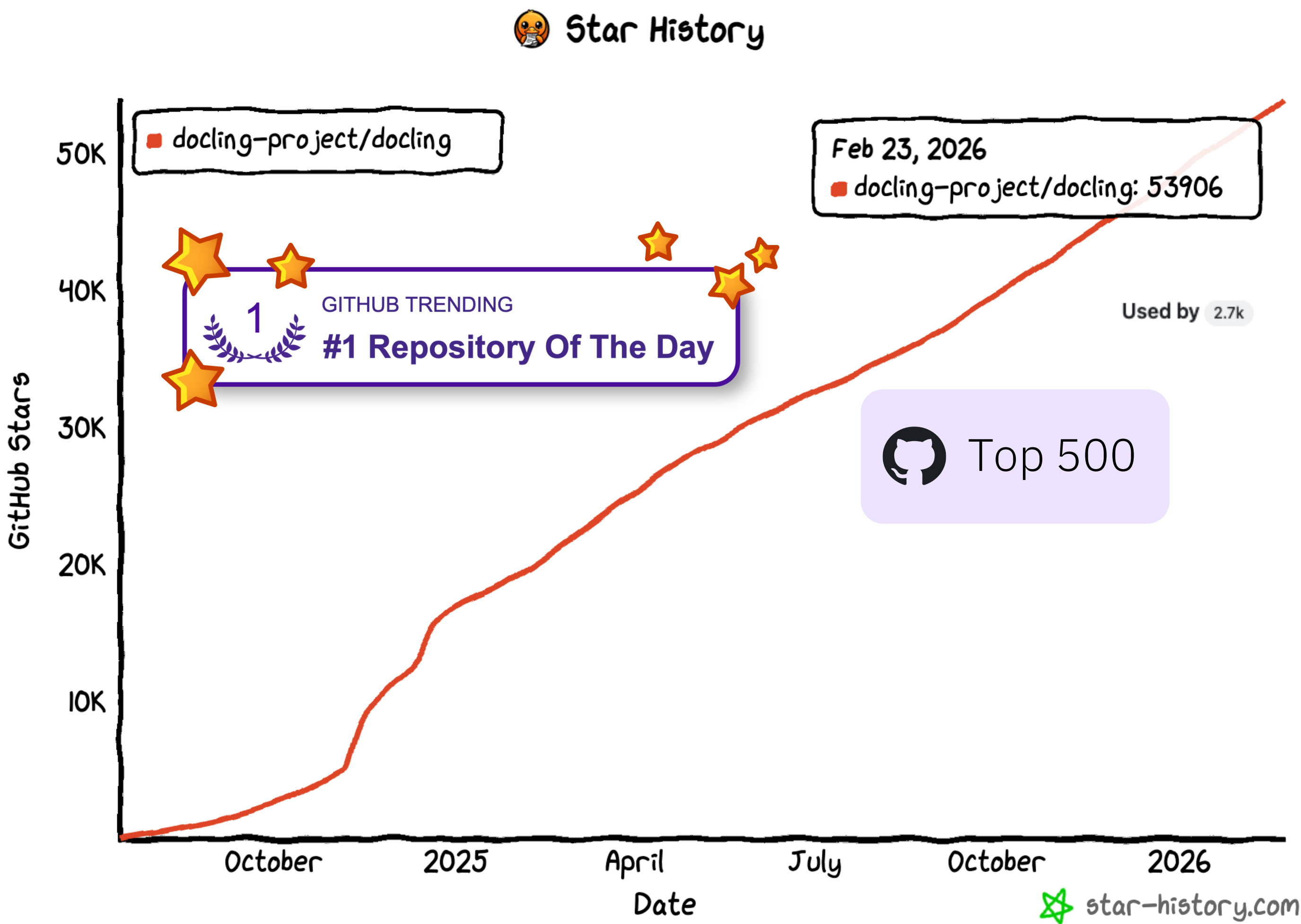
OLFX | Insights

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Community adoption

A rising star in the open-source space

- ★ 50k+ GitHub stars
- 🐍 4M downloads / month on PyPI



Ecosystem

RAG frameworks

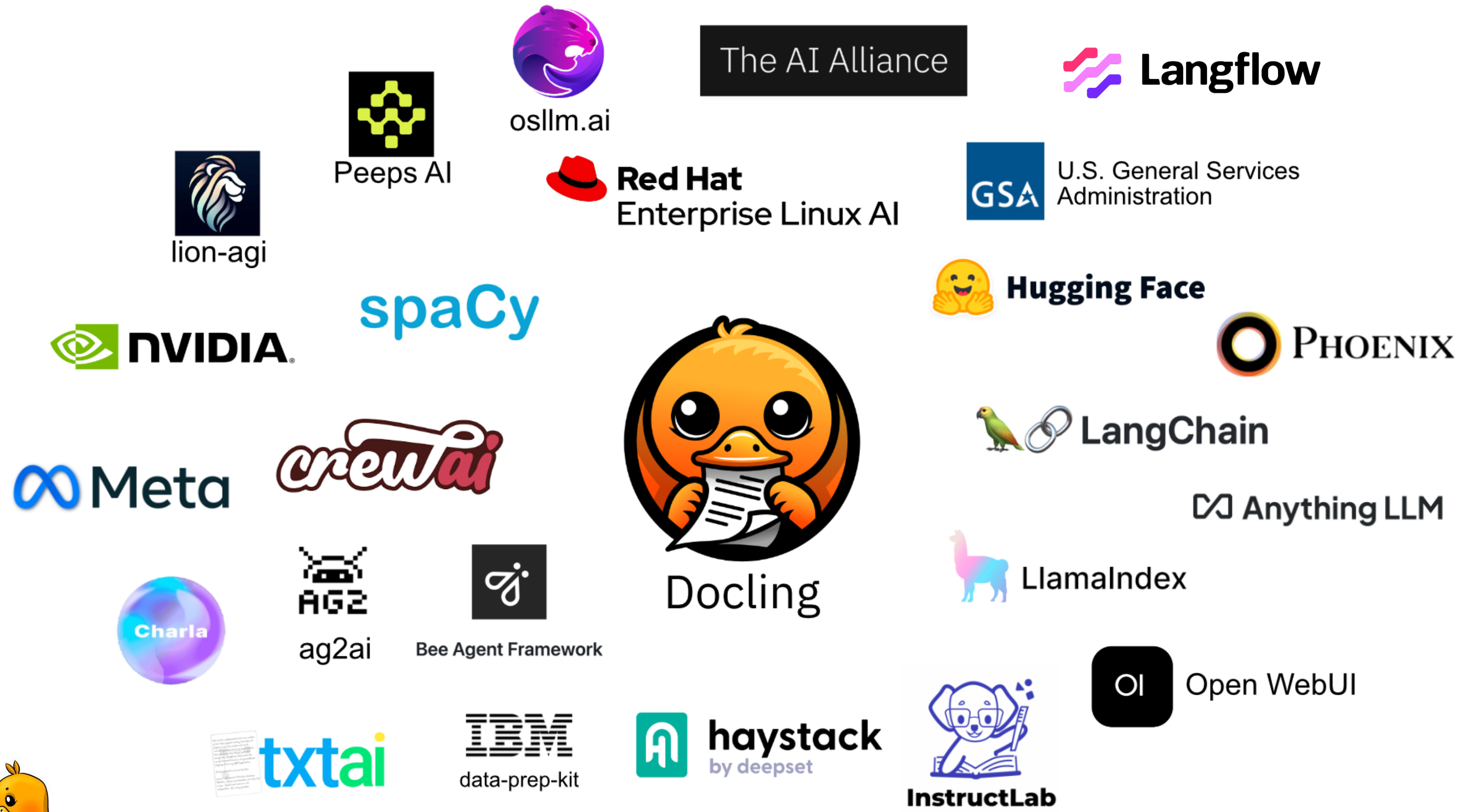
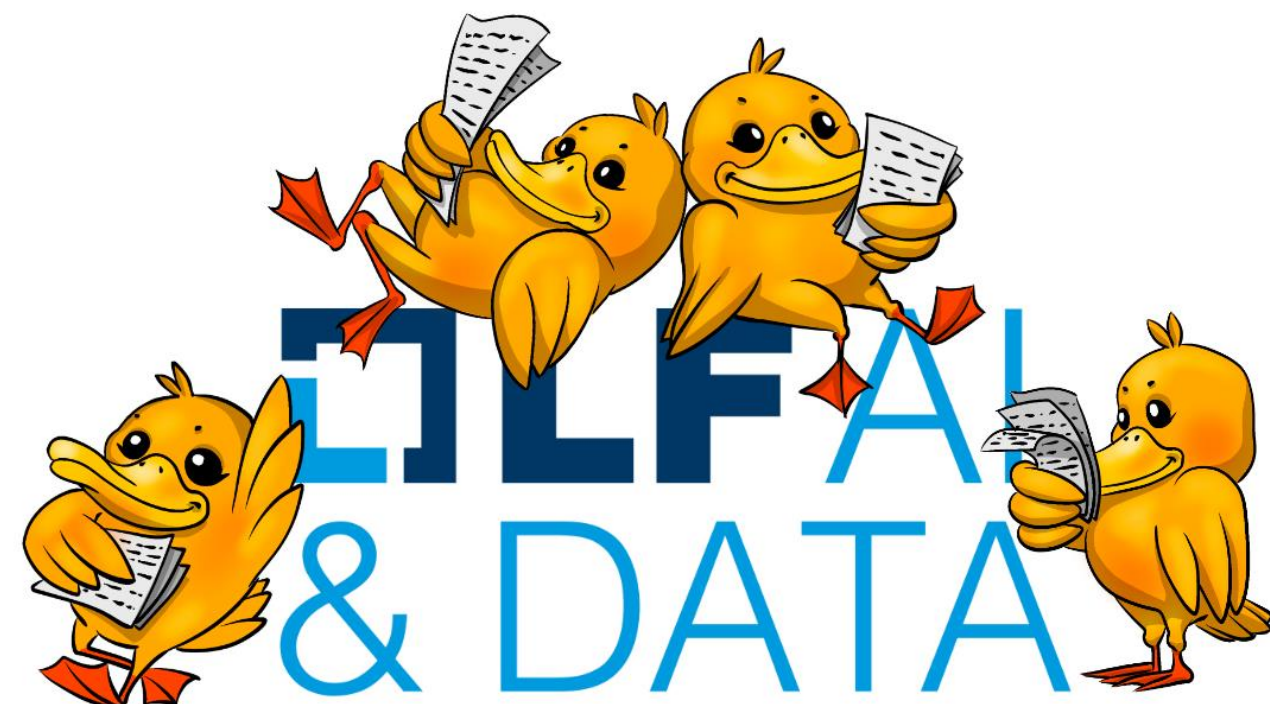
Agentic systems

Dataset generation

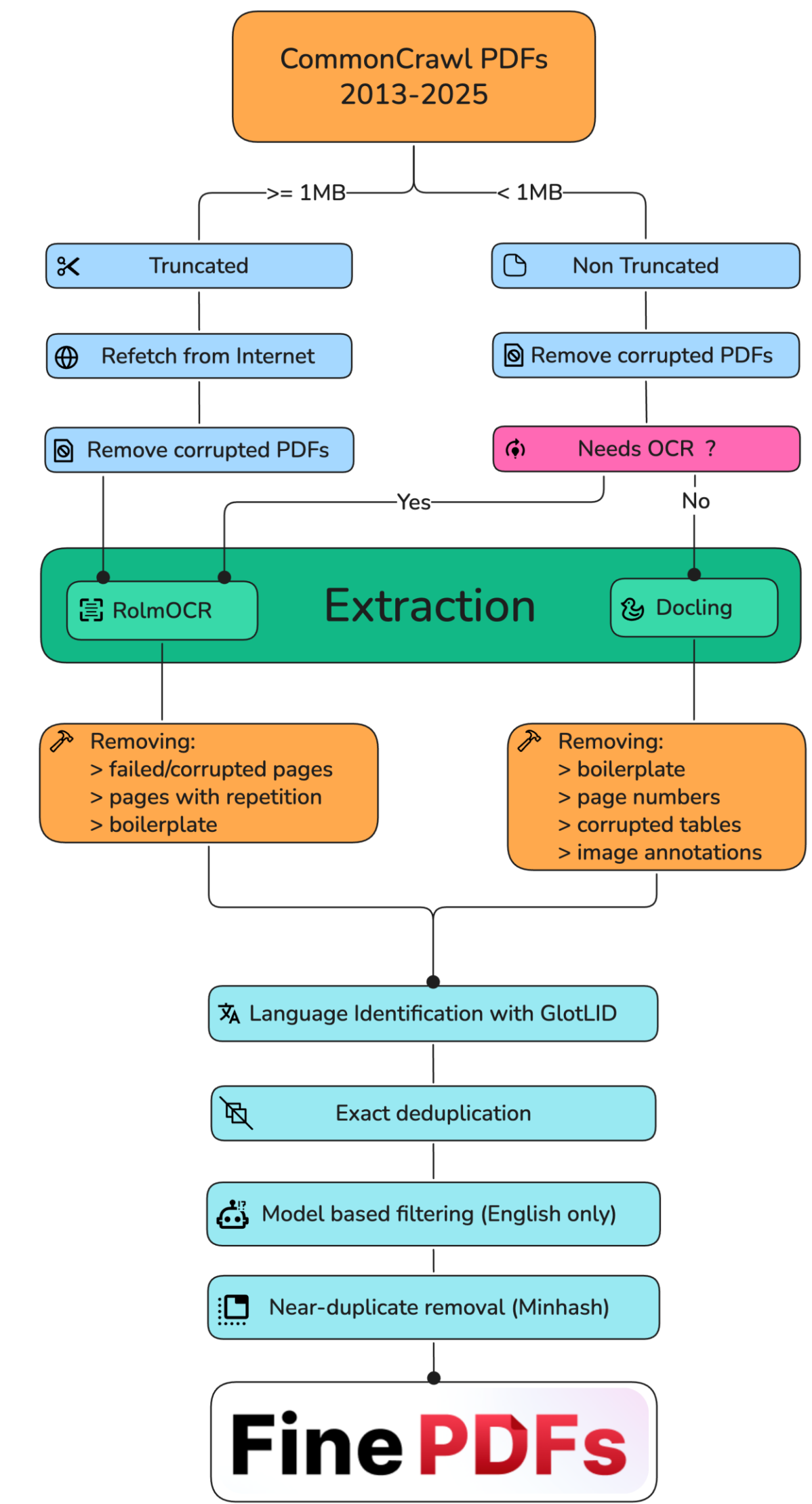
Vector databases

Client solutions

...



Scale, cost, performance



Leandro von Werra • 1st Head of Research at Hugging Face 15h •

The FinePDFs pipeline really shows the massive scale needed for LLM data processing (spoiler: it's ~\$1M):

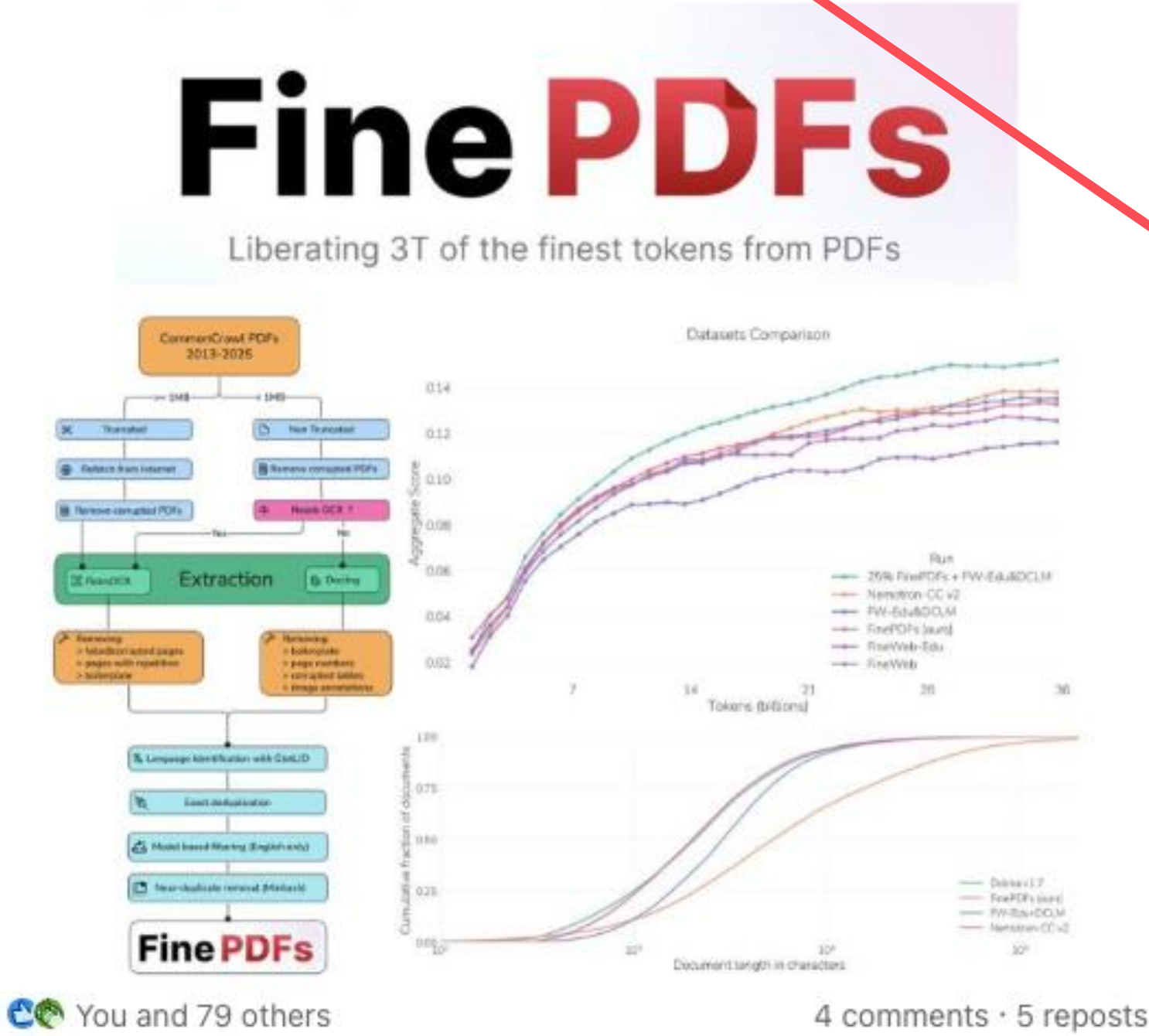
Raw data: 1.35B PDFs (1.2 PB) → \$50k/mo storage

Extraction:
> rolmOCR: 368M docs → 250k GPUh, \$750k
> docling: 918M docs → 2.4M vCPUh, \$35k

Total liberation cost (incl. ablations): ~\$1M

Dataset: <https://lnkd.in/e-muAYtJ>

- 📖 475,019,140 PDFs parsed end-to-end
- 🌐 1,733 languages represented
- 📄 ~3 trillion tokens (~2,918B) extracted
- 💾 3.65 TB of high-quality, deduplicated text
- 📅 Data spanning 2013–2025 across 105 CommonCrawl snapshots



→ $918/368 * 750/35 = 50$

→ Docling 50 times more cost-effective than a VLM

Recovering structured content from PDF

With low-level PDF parsers

! undesired
! page headers

KDD '22, August 14–18, 2022, Washington, DC, USA Birgit Pfitzmann, Christoph Auer, Michele Dolfi, Ahmed S. Nassar, and Peter Staar

Table 1: DocLayNet dataset overview. Along with the frequency of each class label, we present the relative occurrence (as % of row “Total”) in the train, test and validation sets. The inter-annotator agreement is computed as the mAP@0.5-0.95 metric between pairwise annotations from the triple-annotated pages, from which we obtain accuracy ranges.

class label	Count	% of Total			triple inter-annotator mAP @ 0.5-0.95 (%)									
		Train	Test	Val	All	Fin	Man	Sci	Law	Pat	Ten			
Caption	22524	2.04	1.77	2.32	84-89	40-61	86-92	94-99	95-99	69-78	n/a			
Footnote	6318	0.60	0.31	0.58	83-91	n/a	100	62-88	85-94	n/a	82-97			
Formula	25027	2.25	1.90	2.96	83-85	n/a	n/a	84-87	86-96	n/a	n/a			
List-item	185660	17.19	13.34	15.82	87-88	74-83	90-92	97-97	81-85	75-88	93-95			
Page-footer	70878	6.51	5.58	6.00	93-94	88-90	95-96	100	92-97	100	96-98			
Page-header	58022	5.10	6.70	5.06	85-89	66-76	90-94	98-100	91-92	97-99	81-86			
Picture	45976	4.21	2.78	5.31	69-71	56-59	82-86	69-82	80-95	66-71	59-76			
Section-header	142884	12.60	15.77	12.85	83-84	76-81	90-92	94-95	87-94	69-73	78-86			
Table	34733	3.20	2.27	3.60	77-81	75-80	83-86	98-99	58-80	79-84	70-85			
Text	510377	45.82	49.28	45.00	84-86	81-86	88-93	89-93	87-92	71-79	87-95			
Title	5071	0.47	0.30	0.50	60-72	24-63	50-63	94-100	82-96	68-79	24-56			
Total	1107470	941123	99816	66531	82-83	71-74	79-81	89-94	86-91	71-76	68-85			

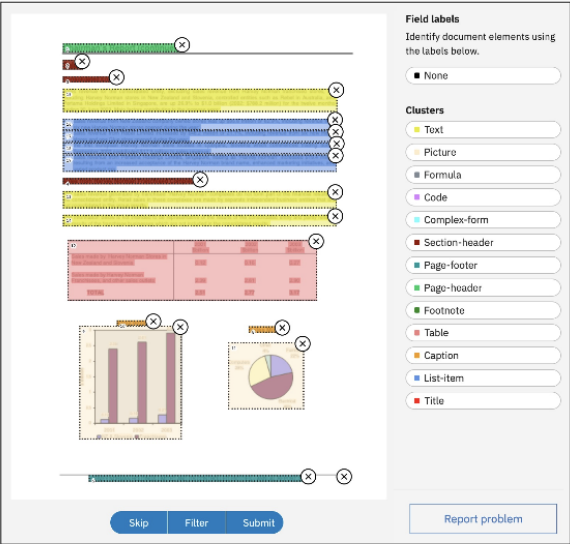


Figure 3: Corpus Conversion Service annotation user interface. The PDF page is shown in the background, with overlaid text-cells (in darker shades). The annotation boxes can be drawn by dragging a rectangle over each segment with the respective label from the palette on the right.

we distributed the annotation workload and performed continuous quality controls. Phase one and two required a small team of experts only. For phases three and four, a group of 40 dedicated annotators were assembled and supervised.

Phase 1: Data selection and preparation. Our inclusion criteria for documents were described in Section 3. A large effort went into ensuring that all documents are free to use. The data sources

include publication repositories such as arXiv³, government offices, company websites as well as data directory services for financial reports and patents. Scanned documents were excluded wherever possible because they can be rotated or skewed. This would not allow us to perform annotation with rectangular bounding-boxes and therefore complicate the annotation process.

Preparation work included uploading and parsing the sourced PDF documents in the Corpus Conversion Service (CCS) [22], a cloud-native platform which provides a visual annotation interface and allows for dataset inspection and analysis. The annotation interface of CCS is shown in Figure 3. The desired balance of pages between the different document categories was achieved by selective subsampling of pages with certain desired properties. For example, we made sure to include the title page of each document and bias the remaining page selection to those with figures or tables. The latter was achieved by leveraging pre-trained object detection models from PubLayNet, which helped us estimate how many figures and tables a given page contains.

Phase 2: Label selection and guideline. We reviewed the collected documents and identified the most common structural features they exhibit. This was achieved by identifying recurrent layout elements and lead us to the definition of 11 distinct class labels. These 11 class labels are *Caption*, *Footnote*, *Formula*, *List-item*, *Page-footer*, *Page-header*, *Picture*, *Section-header*, *Table*, *Text*, and *Title*. Critical factors that were considered for the choice of these class labels were (1) the overall occurrence of the label, (2) the specificity of the label, (3) recognisability on a single page (i.e. no need for context from previous or next page) and (4) overall coverage of the page. Specificity ensures that the choice of label is not ambiguous, while coverage ensures that all meaningful items on a page can be annotated. We refrained from class labels that are very specific to a document category, such as *Abstract* in the *Scientific Articles* category. We also avoided class labels that are tightly linked to the semantics of the text. Labels such as *Author* and *Affiliation*, as seen in DocBank, are often only distinguishable by discriminating on

³<https://arxiv.org/>

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% of Total

triple inter-annotator mAP @ 0.5-0.95 (%)

[...]

Count

22524

6318

25027

185660

70878

58022

45976

142884

34733

510377

5071

1107470

[...]

[...]

include publication repositories such as arXiv³, government offices, company websites as well as data directory services for financial reports and patents. Scanned documents were excluded wherever possible because they can be rotated or skewed. This would not allow us to perform annotation with rectangular bounding-boxes and therefore complicate the annotation process.

[...]

! Multi-column often
! breaks order

! Tables not
! understood

! Image content
! missing

! Line wraps not
! understood

✓ Very fast and cheap

✗ Incomplete

✗ Loss of structure

✗ Noisy

➡ Unfit for most use cases

What about 3rd-party models?

Large models are often challenged too

order to compute the TED score. Inference timing results for all experiments were obtained from the same machine on a single core with AMD EPYC 7763 CPU @2.45 GHz.

5.1 Hyper Parameter Optimization

We have chosen the PubTabNet data set to perform HPO, since it includes a highly diverse set of tables. Also we report TED scores separately for simple and complex tables (tables with cell spans). Results are presented in Table. [1](#). It is evident that with OTSL, our model achieves the same TED score and slightly better mAP scores in comparison to HTML. However OTSL yields a *2x speed up* in the inference runtime over HTML.

Table 1. HPO performed in OTSL and HTML representation on the same transformer-based TableFormer [\[9\]](#) architecture, trained only on PubTabNet [\[22\]](#). Effects of reducing the # of layers in encoder and decoder stages of the model show that smaller models trained on OTSL perform better, especially in recognizing complex table structures, and maintain a much higher mAP score than the HTML counterpart.

# enc-layers	# dec-layers	Language	TEDs			mAP (0.75)	Inference time (secs)
			simple	complex	all		
6	6	OTSL	0.965	0.934	0.955	0.88	2.73
		HTML	0.969	0.927	0.955	0.857	5.39
4	4	OTSL	0.938	0.904	0.927	0.853	1.97
		HTML	0.952	0.909	0.938	0.843	3.77
2	4	OTSL	0.923	0.897	0.915	0.859	1.91
		HTML	0.945	0.901	0.931	0.834	3.81
4	2	OTSL	0.952	0.92	0.942	0.857	1.22
		HTML	0.944	0.903	0.931	0.824	2

5.2 Quantitative Results

We picked the model parameter configuration that produced the best prediction quality (enc=6, dec=6, heads=8) with PubTabNet alone, then independently trained and evaluated it on three publicly available data sets: PubTabNet (395k samples), FinTabNet (113k samples) and PubTables-1M (about 1M samples). Performance results are presented in Table. [2](#) It is clearly evident that the model trained on OTSL outperforms HTML across the board, keeping high TEDs and mAP scores even on difficult financial tables (FinTabNet) that contain sparse and large tables.

Additionally, the results show that OTSL has an advantage over HTML when applied on a bigger data set like PubTables-1M and achieves significantly improved scores. Finally, OTSL achieves faster inference due to fewer decoding steps which is a result of the reduced sequence representation.

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6	6	HTML	0.969	0.927	0.955
4	4	OTSL	0.938	0.904	0.927
4	4	HTML	0.952	0.909	0.938
2	2	OTSL	0.923	0.897	0.915
2	2	HTML	0.945	0.901	0.931
4	2	OTSL	0.952	0.92	0.942
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Optimized Table Tokenization for Table Structure Recognition

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Quality

Low quality = poor results



gurovdigital 15 h

lol, over 20 scientific papers now feature the

were incubated with an extract from spores and integrated at pH 7.0. Peptide was released which established that the coats contained substrate for the lytic enzyme present in spores. Peptide was also released from spore coats of *B. megaterium* by the action of the enzyme from *B. cereus* spores. The lytic enzyme did not attack intact resting spores.

The spore develops in the vegetative cell, which thus becomes a sporangium. It is by no means certain what happens to the vegetative cell wall when the spore is released. In *Clostridium* species it appears that at least part of this structure is retained as an outer membrane around the spore. It is the opinion of some workers that the wall of the sporulating cell forms the exosporium which exists as an outer

characteristic type. It was concluded that at least part of the sporangial wall was dissolved away to allow release of the spore. It appears likely that the exosporium of *B. cereus* does not have a composition similar to that of the vegetative cell wall, from the results obtained by Dr. J. R.

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Date syrup (as one of the agricultural wastes) was used to produce bacterial cellulose using *Gluconastobacter xylinus*. Fourier transform infrared spectroscopy (FTIR), vegetative electron microscopy, and X-ray diffraction were used to determine the structure of bacterial cellulose, cellulose fibers, and crystallinity of the samples (Moosavi and

Silver and gold nanoparticles for

[HTML] m



692

12

43



BRONZE PIECES FROM JEYRAN TEPE, UZBAKI
B. SODAEI, H. RAHNEMA - researchgate.net
This study is a report of the results of metallographic study of
5 bronze pieces found in Jeyrān Tepe dating back to the Iron

Quality

High quality = trust

How many PDF pages does the DocLayNet dataset contain?

DocLayNet contains 80863 PDF pages.



Text Location - Page 3

KDD '22, August 14–18, 2022, Washington, DC, USA Birgit Pfitzmann, Christoph Auer, Michele Dolfi, Ahmed S. Nassar, and Peter Staar

1 INTRODUCTION

Despite the substantial improvements achieved with machine-learning (ML) approaches and deep neural networks in recent years, document conversion remains a challenging problem, as demonstrated by the numerous public competitions held on this topic [1–4]. The challenge originates from the huge variability in PDF documents regarding layout, language and formats (scanned, programmatic or a combination of both). Engineering a single ML model that can be applied on all types of documents and provides high-quality layout segmentation remains to this day extremely challenging [5]. To highlight the variability in document layouts, we show a few example documents from the DocLayNet dataset in Figure 1.

A key problem in the process of document conversion is to understand the structure of a single document page, i.e. which segments of text should be grouped together in a unit. To train models for this task, there are currently two large datasets available to the community, PubLayNet [6] and DocBank [7]. They were introduced in 2019 and 2020 respectively and significantly accelerated the implementation of layout detection and segmentation models due to their sizes of 300K and 500K ground-truth pages. These sizes were achieved by leveraging an automation approach. The benefit of automated ground-truth generation is obvious: one can generate large ground-truth datasets at virtually no cost. However, the automation introduces a constraint on the variability in the dataset, because corresponding structured source data must be available. PubLayNet and DocBank were both generated from scientific document repositories (PubMed and arXiv), which provide XML or $\LaTeX$ sources. Those scientific documents present a limited variability in their layouts, because they are typeset in uniform templates provided by the publishers. Obviously, documents such as technical manuals, annual company reports, legal text, government tenders, etc. have very different and partially unique layouts. As a consequence, the layout predictions obtained from models trained on PubLayNet or DocBank is very reasonable when applied on scientific documents. However, for more *artistic* or *free-style* layouts, we see sub-par prediction quality from these models, which we demonstrate in Section 5.

In this paper, we present the DocLayNet dataset. It provides page-by-page layout annotation ground-truth using bounding-boxes for 11 distinct class labels on 80863 unique document pages, of which a fraction carry double- or triple-annotations. DocLayNet is similar in spirit to PubLayNet and DocBank and will likewise be made available to the public¹ in order to stimulate the document-layout analysis community. It distinguishes itself in the following aspects:

- (1) *Human Annotation*: In contrast to PubLayNet and DocBank, we relied on human annotation instead of automation approaches to generate the data set.
- (2) *Large Layout Variability*: We include diverse and complex layouts from a large variety of public sources.
- (3) *Detailed Label Set*: We define 11 class labels to distinguish layout features in high detail. PubLayNet provides 5 labels; DocBank provides 13, although not a superset of ours.
- (4) *Redundant Annotations*: A fraction of the pages in the DocLayNet data set carry more than one human annotation.

This enables experimentation with annotation uncertainty and quality control analysis.

(5) *Pre-defined Train-, Test- & Validation-set*: Like DocBank, we provide fixed train-, test- & validation-sets to ensure proportional representation of the class-labels. Further, we prevent leakage of unique layouts across sets, which has a large effect on model accuracy scores.

All aspects outlined above are detailed in Section 3. In Section 4, we will elaborate on how we designed and executed this large-scale human annotation campaign. We will also share key insights and lessons learned that might prove helpful for other parties planning to set up annotation campaigns.

In Section 5, we will present baseline accuracy numbers for a variety of object detection methods (Faster R-CNN, Mask R-CNN and YOLOv5) trained on DocLayNet. We further show how the model performance is impacted by varying the DocLayNet dataset size, reducing the label set and modifying the train/test-split. Last but not least, we compare the performance of models trained on PubLayNet, DocBank and DocLayNet and demonstrate that a model trained on DocLayNet provides overall more robust layout recovery.

2 RELATED WORK

While early approaches in document-layout analysis used rule-based algorithms and heuristics [8], the problem is lately addressed with deep learning methods. The most common approach is to leverage object detection models [9–15]. In the last decade, the accuracy and speed of these models has increased dramatically. Furthermore, most state-of-the-art object detection methods can be trained and applied with very little work, thanks to a standardisation effort of the ground-truth data format [16] and common deep-learning frameworks [17]. Reference data sets such as PubLayNet [6] and DocBank provide their data in the commonly accepted COCO format [16].

Lately, new types of ML models for document-layout analysis have emerged in the community [18–21]. These models do not approach the problem of layout analysis purely based on an image representation of the page, as computer vision methods do. Instead, they combine the text tokens and image representation of a page in order to obtain a segmentation. While the reported accuracies appear to be promising, a broadly accepted data format which links geometric and textual features has yet to establish.

3 THE DOCLAYNET DATASET

DocLayNet contains 80863 PDF pages. Among these, 7059 carry two instances of human annotations, and 1591 carry three. This amounts to 91104 total annotation instances. The annotations provide layout information in the shape of labeled, rectangular bounding-boxes. We define 11 distinct labels for layout features, namely *Caption*, *Footnote*, *Formula*, *List-Item*, *Page-footer*, *Page-header*, *Picture*, *Section-header*, *Table*, *Text*, and *Title*. Our reasoning for picking this particular label set is detailed in Section 4.

In addition to open intellectual property constraints for the source documents, we required that the documents in DocLayNet adhere to a few conditions. Firstly, we kept scanned documents

¹<https://developer.ibm.com/exchanges/data/all/doclaynet>

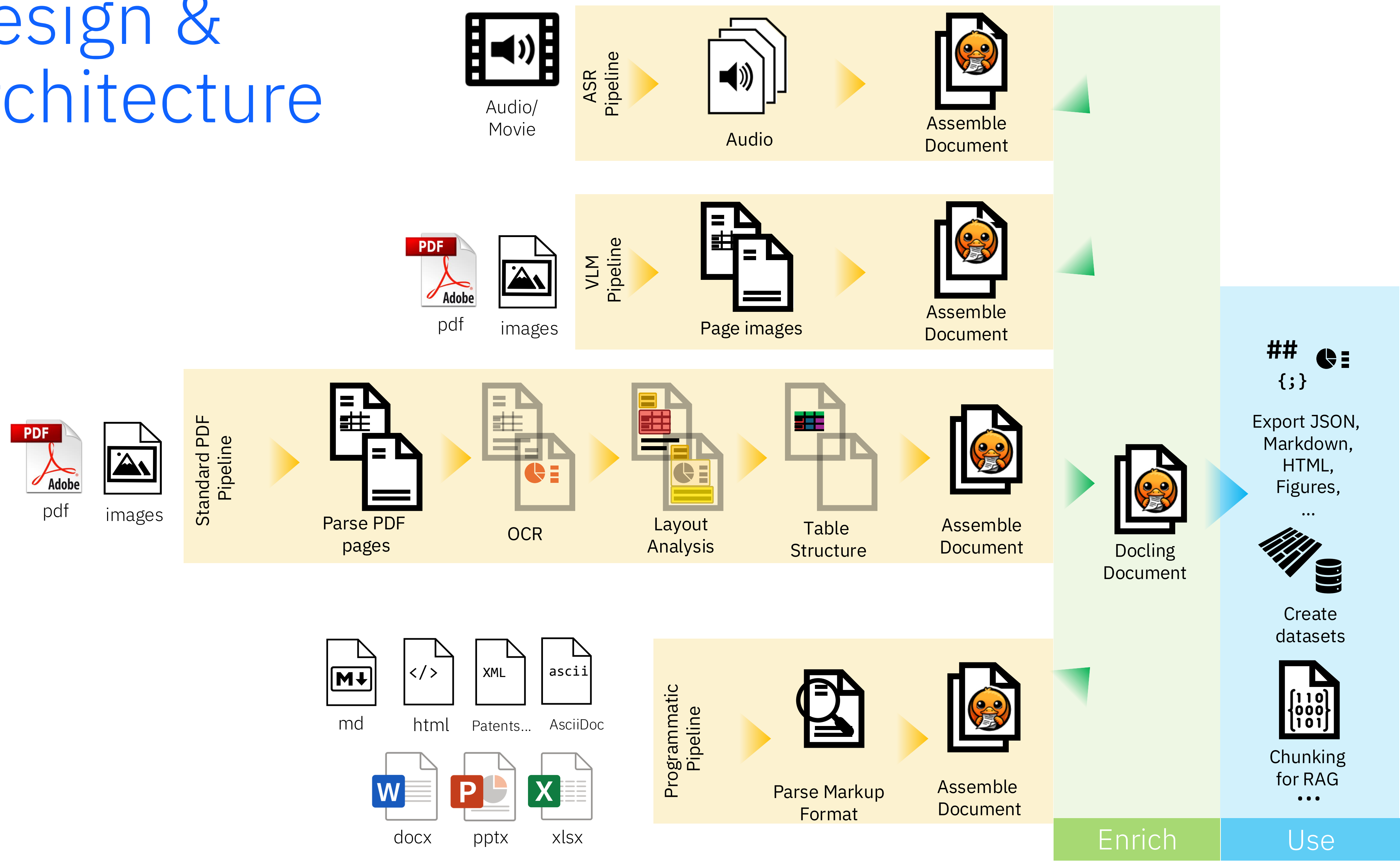
RAG with visual grounding



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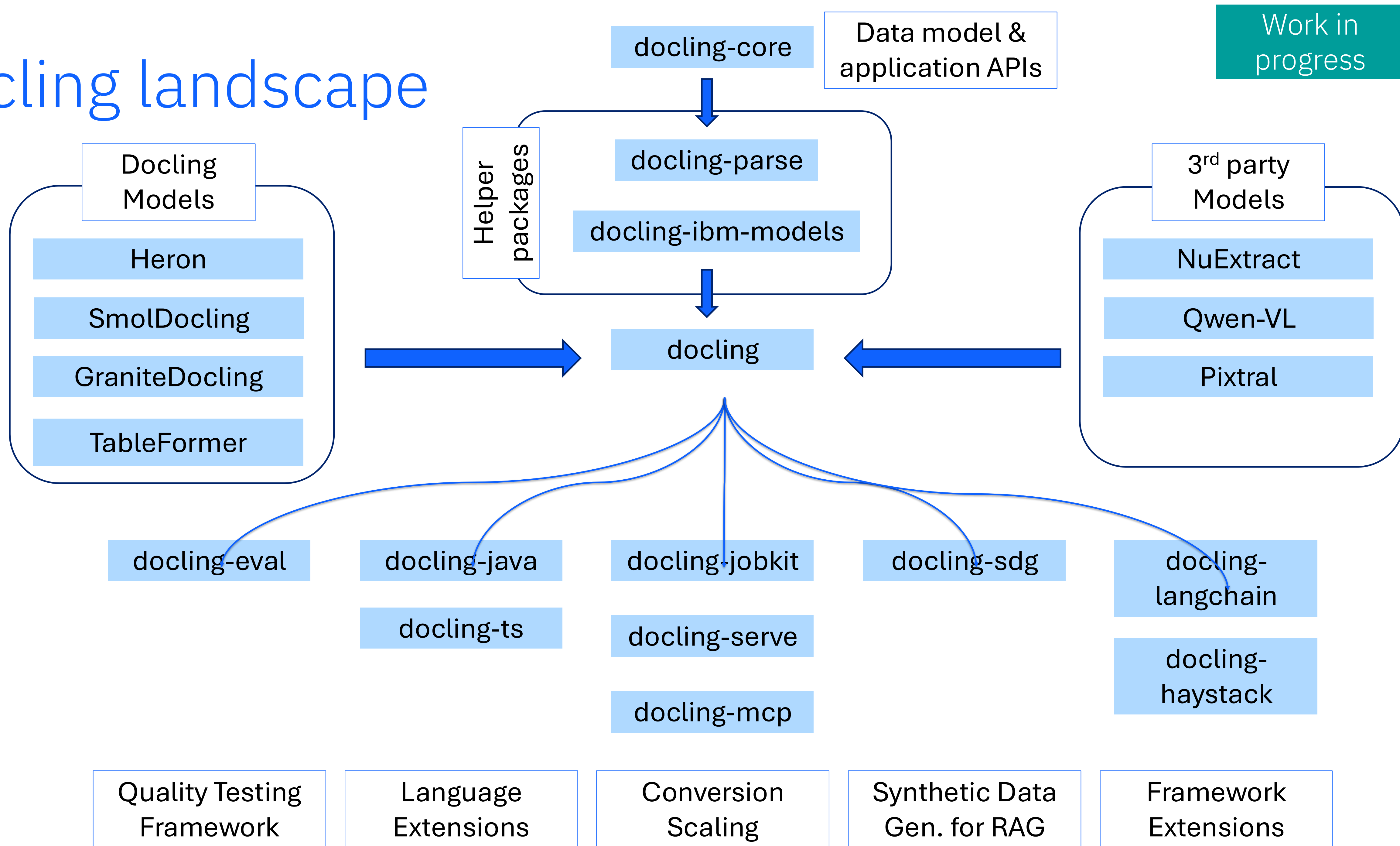
Deep dive

Design & architecture



Docling landscape

Work in progress



A new VLM generation

Ultra-small, built on specialized, LLM-friendly tokens

SmolDocling

Single VLM Model
for OCR / layout /
table / charts

Granite Docling

Enhanced stability
Production quality
Multi-lingual

Mar 17



Trending on 🧡 this week

Models

ds4sd/SmolDocling-256M-preview

Updated 1 day ago • 📄 27.9k • ❤️ 847

mistralai/Mistral-Small-3.1-24B-Ins...

Updated 2 days ago • 📄 60.4k • ❤️ 924

manycore-research/SpatialLM-Llama-1B

Updated 4 days ago • 📄 2.38k • ❤️ 588

sesame/csm-1b

Updated 8 days ago • 📄 32k • ❤️ 1.58k

deepseek-ai/DeepSeek-V3-0324

Updated about 6 hours ago • ❤️ 471

[Browse 1M+ models](#)



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+950k

Sep 17



Clem Delangue 🧡🌐 • Following

Co-founder & CEO at Hugging Face

1w • 🌐

IBM is doing the grand slam of trending (model, dataset & space) on [Hugging Face](#) for projects they either led (Docling) or contributed to (finepdfs). ...more



Trending on 🧡 this week

Models

ibm-granite/granite-docling-256M

Updated about 21 hours ago • 📄 25.7k • ❤️ 585

openbmb/VoxCPM-0.5B

Updated 5 days ago • 📄 3.24k • ❤️ 649

Alibaba-NLP/Tongyi-DeepResearch-30B-A3B

Updated 7 days ago • 📄 9.53k • ❤️ 461

Wan-AI/Wan2.2-Animate-14B

Updated 5 days ago • 📄 14.4k • ❤️ 379

Qwen/Qwen3-Demo-30B-A3B-Instruct

Updated 1 day ago • 📄 1.49k • ❤️ 313

[Browse 1M+ models](#)

Spaces

DeepSite v2.0

Generate any application with DeepSite

Wan2.2 Animate

Wan2.2 14B Fast

granite-docling-256M demo

Generate images to structured documents and answer questions

IndexTTS 2 Demo

Generate expressive speech from text with emotion control

[Browse 400k+ applications](#)

Datasets

HuggingFaceW/FinepdfIs

Updated 18 days ago • 📄 95.2k • ❤️ 364

fka/awesome-chatgpt-prompts

Updated Jan 6 • 📄 48.4k • ❤️ 9.12k

IntzenRobotics/Omniverse

Updated 24 minutes ago • 📄 15.2k • ❤️ 59

LucasFang/FLUX-Reason-6M

Updated 12 days ago • 📄 38.5k • ❤️ 78

HuggingFaceW/FineVision

Updated 19 days ago • 📄 257k • ❤️ 342

[Browse 250k+ datasets](#)

👍❤️ You and 880 others

31 comments · 111 reposts



$$\int^{\lambda_{i+1}} B(\lambda, T_{i-1}) d\lambda = 1$$

Schema-based information extraction

Extracting specific pieces of information from a document

Eventure
EVENT PLANNER

hello

THIS IS YOUR INVOICE

BILLED TO: Jonathan Patterson
PAY TO: Liceria & Co.
123 Anywhere St., Any City
Bank: Borcele Bank
Account No: 0123 4567 8901

DESCRIPTION	UNIT PRICE	QTY	TOTAL
Flower Arrangements	100	1	\$100
Ballroom Planning	100	1	\$100
Wedding Hall Decor	100	1	\$100
Reception Planning	100	1	\$100
Concierge	100	1	\$100
SUBTOTAL			\$500
			Tax 10%
TOTAL			\$550

INVOICE NO: 01234
DATE: 11.02.2030
DUE DATE: 11.03.2030

THANK YOU

+

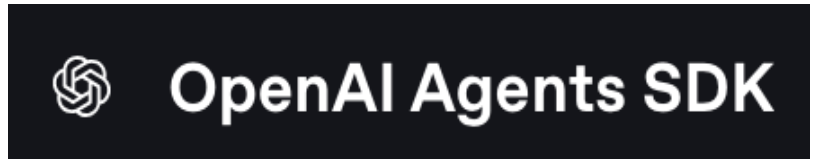
```
invoice_dict = {  
    "bill number": "string",  
    "total invoice price": "float",  
    "currency of total invoice price": "string",  
    "name of invoice addressee": "string",  
    "name of invoice sender": "string",  
}
```



```
{  
    'bill number': '01234',  
    'total invoice price': 550,  
    'currency of total invoice price': 'USD',  
    'name of invoice addressee': 'Jonathan Patterson',  
    'name of invoice sender': 'Eventure Event Planner'  
}
```

Powering AI agents

- **Agent** = AI application that can act autonomously to understand, plan & execute a task
- MCP = protocol for connecting AI applications to external tools & data sources
- Docling provides an MCP server with tools for document conversion, manipulation & content creation
- Plug & play on multiple frameworks & AI desktop clients



Further recent highlights

Docling Java project



NVIDIA x Docling collab

 **NVIDIA DEVELOPER** Home Blog Forums Docs Downloads Training

Technical Blog

An Agentic AI toolkit for local AI

The use cases for private, local agents are endless. But building reliable, repeatable, and high-quality private agents remains a challenge. LLM quality deteriorates when you distill and quantize the model to fit within a limited VRAM budget on PC. The need for accuracy increases as agentic workflows require reliable and repeatable answers when interfacing with other tools or actions.

To address this, developers typically use two tools to increase accuracy: fine-tuning and retrieval-augmented-generation (RAG). NVIDIA released updates to accelerate tools across this workflow for building agentic AI.

Nemotron 3 Nano is a 32B parameter MoE model optimized for agentic AI and fine-tuning. With 3.6B active parameters and a 1M context window, it tops several benchmarks across coding, instruction-following, long-context reasoning, and STEM tasks. The model is optimized for RTX PCs and DGX Spark via [Ollama](#) and [llama.cpp](#), and can be fine-tuned using [Unsloth](#).

This model stands out for being the most open, with weights, recipes, and datasets widely available. Open models and datasets make customizing the model easier for developers. They prevent redundant fine-tuning and eliminate data leakage for objective benchmarking for robust and efficient workflows. [Get started](#) with LoRA-based fine-tuning for it.

For RAG, NVIDIA partnered with [Docling](#)—a package to ingest, analyze, and process documents into a machine-understandable language for RAG pipelines. Docling is optimized for RTX PCs and DGX Spark and delivers 4x performance compared to CPUs.

There are two ways of using Docling:

1. Traditional OCR pipeline: This is a pipeline of libraries and models that is accelerated via PyTorch-CUDA on RTX.
2. VLM-based pipeline: An advanced pipeline for complex multi-modality documents, available for use via vLLM within WSL and Linux environments.

Docling is developed at IBM and contributed to the Linux Foundation. Start now on RTX with this [easy-to-use guide](#).

LaTeX parsing



Docling Now Natively Ingests LaTeX

Academics: Start processing your LaTeX corpus to make it AI-ready!

github.com/docling-project/docling

- ✓ Image Referencing
 - `\includegraphics \`
 - Figure 1: An example image
- ✓ Complex Table Parsing
 - Parameter | Value 1
 - 1.309 | 1.30
 - 0.106 | 19.4
 - 0.109 | 1.5
- ✓ Inline Math & Equations
 - `E=mc^2`
 - `\praselec{8{8aning +}7,`
 - `<a>->)/g`
 - `(+-->$6ling=0>)`
 - Value 2

Chart understanding



Year	Sales (K)	Growth (%)
2018	35	5%
2019	52	12%
2020	78	18%
2021	89	14%
2022	105	20%

From Chart to Data!

Powered by **IBM Granite**

Roadmap

2026H1 Roadmap

Document ISO Standard

- DocTags as a shared standard
- Focus on compatibility with LLM tokenizers
- Represent complex, multimodal content with visual grounding

New models

- New picture classification
improved quality
better performance
- New table model
new architecture
high-throughput batching
- New GraniteDocling
Two-stages prediction
Variable-resolution patching
Flock of LLMs

Docling Eval

- Simple-to-use and open evaluation suite
- Evaluation metrics tuned for documents
- Complete annotated datasets with all features
- Compatibility and comparison with other evaluation and metrics

ISO standard through LF AI & Data

Motivation

The motivation for this new markup language is twofold,

- 1. It is created from the ground up to be able to represent complex, multimodal content with visual grounding in plain text with markup
- 2. It is created with the express purpose to be compatible with LLM tokenizers, i.e. use a markup structure that maps naturally (== a 1-to-1 mapping between DocTags tokens and LLM tokens) and efficiently (== minimal token count).

▼ Use of form headings

Alien's Change of Address Card

USCIS
Form AR-11
OMB No. 1615-0007
Expires 08/31/2024

NOTE: An asterisk (*) indicates a mandatory field that must be completed.

Information About You

*Family Name (Last Name)

starr

*Given Name (First Name)

Peter

Middle Name (if applicable)

WJ

I am in the United States as a:

☐ Visitor

☐ Student

☐ Permanent Resident

☐ Other (Specify)

Country of Citizenship

*Date of Birth (mm/dd/yyyy)

Alien Registration Number (A-Number) (if any)

▶ A-

Information About Your Address

*Present Physical Address (No PO Boxes)

*Street Number and Name

Apt. Ste. Flr.

☐

☐

☐

Number

*City or Town

*State

*ZIP Code

Previous Physical Address

Street Number and Name

Apt. Ste. Flr.

☐

☐

☐

Number

City or Town

State

ZIP Code

A

	Observer 1			
	benign	malignant	Total	
Observer 2	benign	13	2	15
	malignant	0	62	62
Total observer 1	13	64	77	

B

C

HTML

sequence length: 55

<table>

<tr>

<td colspan="2" rowspan="2">

</td>

<td colspan="3" rowspan="2">

</td>

</tr>

<td>

</td>

<td>

</td>

<td>

</td>

</tr>

<td rowspan="3">

</td>

<td>

</td>

<td>

</td>

<td>

</td>

<td>

</td>

</tr>

<td>

</td>

<td>

</td>

<td>

</td>

<td>

</td>

</tr>

</table>

vocabulary for this table: 12 tokens

<table> <tr> </tr> <td> </td> <td colspan="2" colspan="3"> </table>

D

OTSL

sequence length: 30

C

L

C

L

L

NL

U

X

C

C

C

NL

C

C

C

C

C

NL

U

C

C

C

C

NL

U

C

C

C

C

NL

vocabulary: 5 tokens

C

L

U

X

NL

```
<form>
  <form_heading>Information about you</form_heading>
  <form_item>
    <key>
      \*Family Name (Last Name)
    </key>
    <value>starr</value>
  </form_item>
  <form_item>
    <key>\*Given Name (First Name)</key>
    <value>peter</value>
  </form_item>
  <form_item>
    <key>\*Middle Name (if applicable)</key>
    <value>WJ</value>
  </form_item>
  <form_item>
    <key>I am in the United States as a:</key>
    <checkbox selected="false">Visitor</checkbox>
    <checkbox selected="false">Student</checkbox>
    <checkbox selected="false">Permanent Resident</checkbox>
    <checkbox selected="false">Other (Specify)</checkbox>
    <value></value>
  </form_item>
```

Original table:

		2025 Goals			
Continent	Country	2025 Goals	2025 Goals	2025 Goals	2025 Goals
Asia	Japan	45,000	10,000	10,000	10,000
Europe	France	50,000	10,000	10,000	10,000
Europe	Germany	55,000	10,000	10,000	10,000
	Italy	60,000	10,000	10,000	10,000
North America	Canada	65,000	10,000	10,000	10,000
	United States	70,000	10,000	10,000	10,000

Table split across pages:

Page 1

		2025 Goals	
Continent	Country	2025 Goals	2025 Goals
Asia	Japan	45,000	10,000

Page 2

		2025 Goals	
Europe	France	50,000	10,000
Germany	55,000	10,000	10,000

Page 3

		2025 Goals	
Italy	60,000	10,000	10,000
Canada	65,000	10,000	10,000

Page 4

		2025 Goals	
United States	70,000	10,000	10,000

Table threads:

h_thread_1

		2025 Goals	
Continent	Country	2025 Goals	2025 Goals
Asia	Japan	45,000	10,000

h_thread_2

		2025 Goals	
Europe	France	50,000	10,000
Germany	55,000	10,000	10,000

h_thread_3

		2025 Goals	
Italy	60,000	10,000	10,000
Canada	65,000	10,000	10,000

h_thread_4

		2025 Goals	
United States	70,000	10,000	10,000

The scenario in the above figure is represented below.

- We introduce a new horizontal thread token, `h_thread_N`, which is used to capture table content that spans pages sidewise, similarly to the usual thread tokens `thread_N`.
- Only the content that is visible within the page is included in the OTSL token (e.g. see "2025 d").
- When thread linking is resolvable through `u cel / l cel` or `h_thread_N`, the `thread_N` token is not used, as it would be redundant.
- When thread linking must be captured, we capture it the earliest possible, i.e. we don't wait for the bottom-most cell to be reached to add the thread for "Europe" in the example above.

```
...
<!-- page 1: -->
<table>
  <otsl><loc_x0/><loc_y0/><loc_x1/><loc_y1/>
  <thread_1/>
  <h_thread_1/>

  <ecel/>
  <ecel/>
  <rhed/><thread_2/>
</otsl>
</table>
<page_break/>
```

<ecel/>

<ched/>Continent

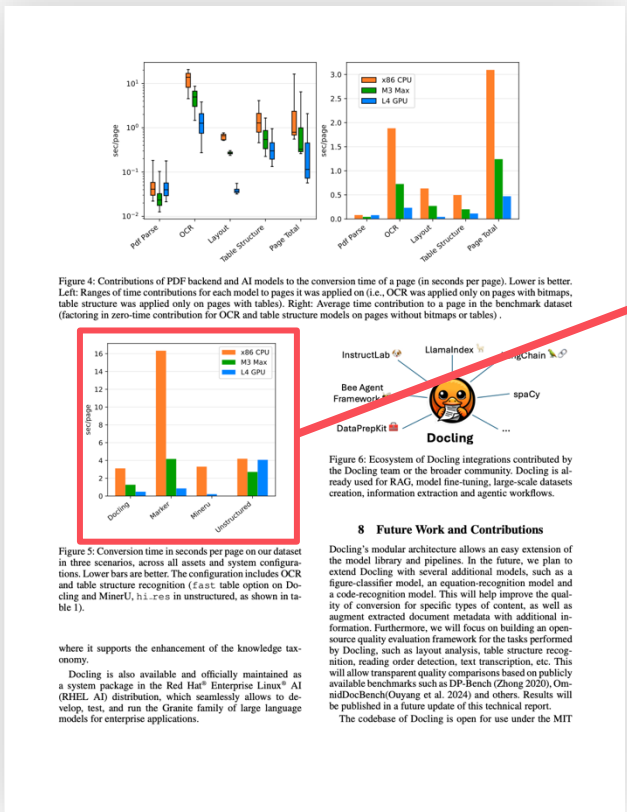
<rhed/>Asia

New models

Standard pipeline

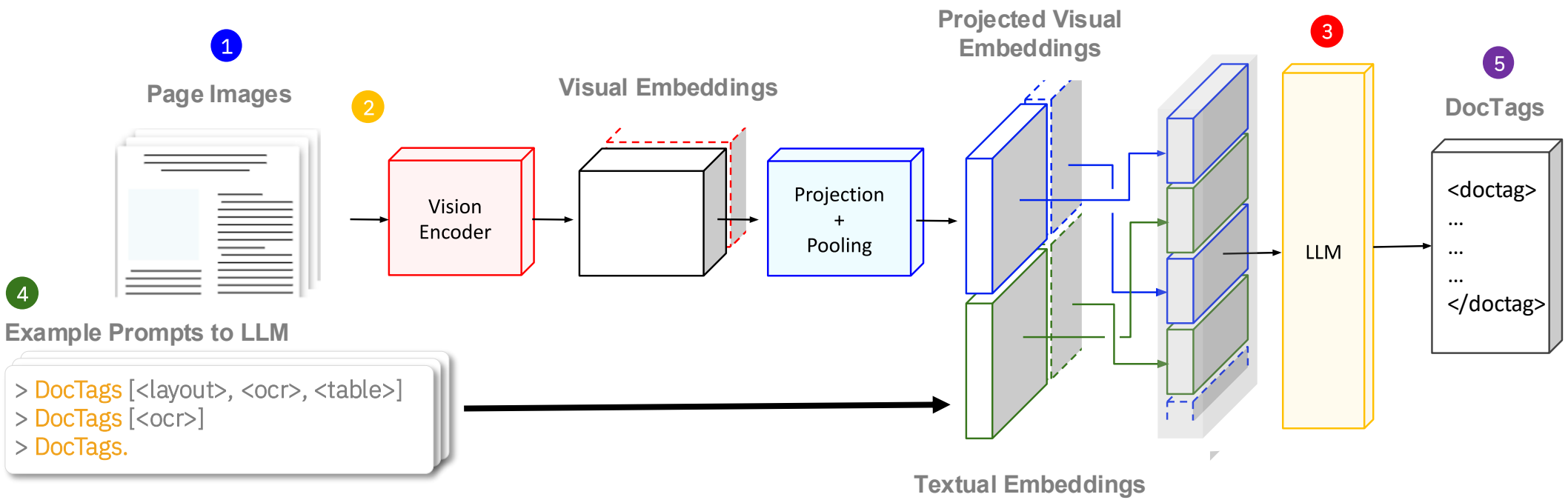
- New picture classification
improved quality
better performance
- New table model
new architecture
high-throughput batching

PictureClassificationClass(
class_name='bar_chart',
confidence=0.99995
)



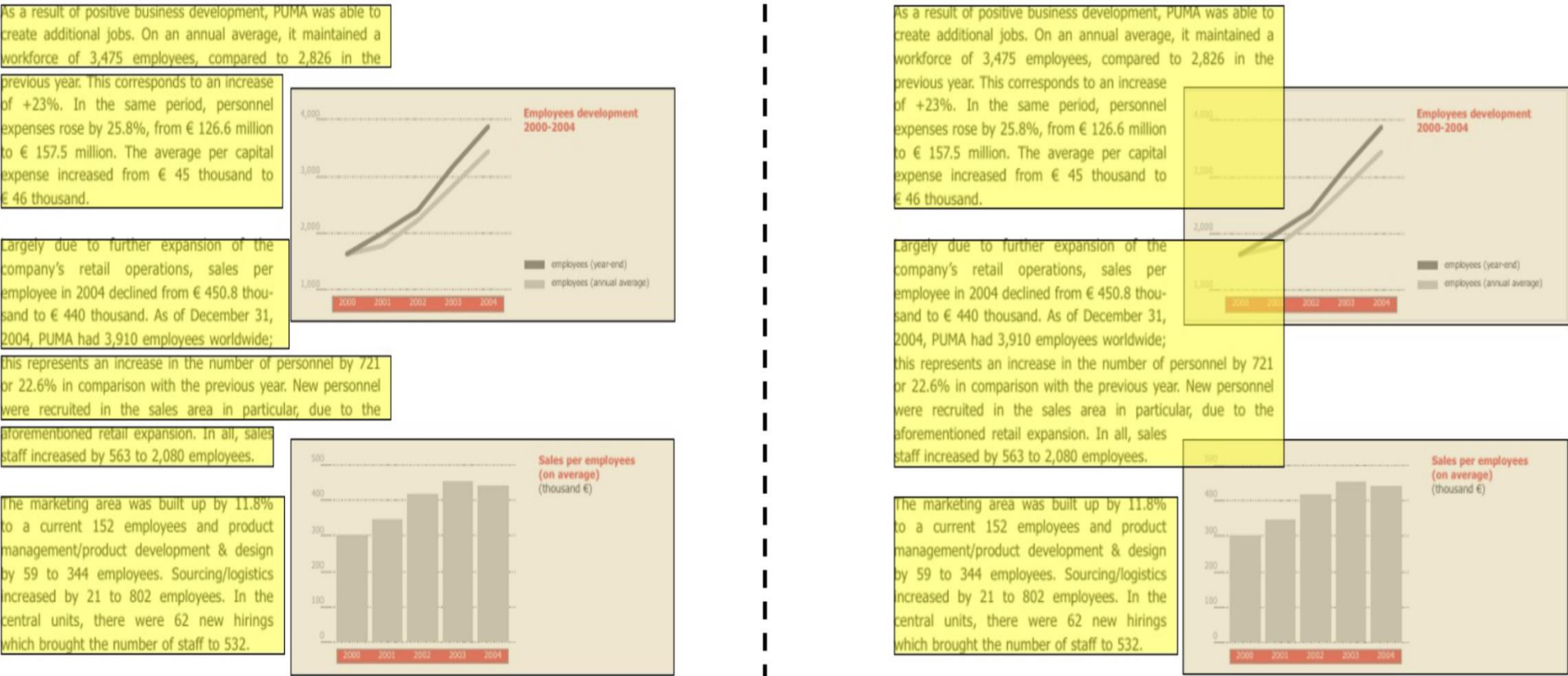
VLM pipeline – New Granite Docling

- 1 Adaptive Pixel-Shuffle Compression: High resolution where it matters, aggressively compressed where it doesn't
- 2 Variable-resolution patching: Preserves fine detail while reducing compute on large or irregular images
- 3 Flock of LLMs: Granite 4 (350M, 1B, 3B*) operating collaboratively
- 4 Custom Conversion Prompts: Enables training without full ground truth and drives precise structured outputs instead of full responses
- 5 ISO DocTags: Standardized document markup for consistent structure and downstream processing

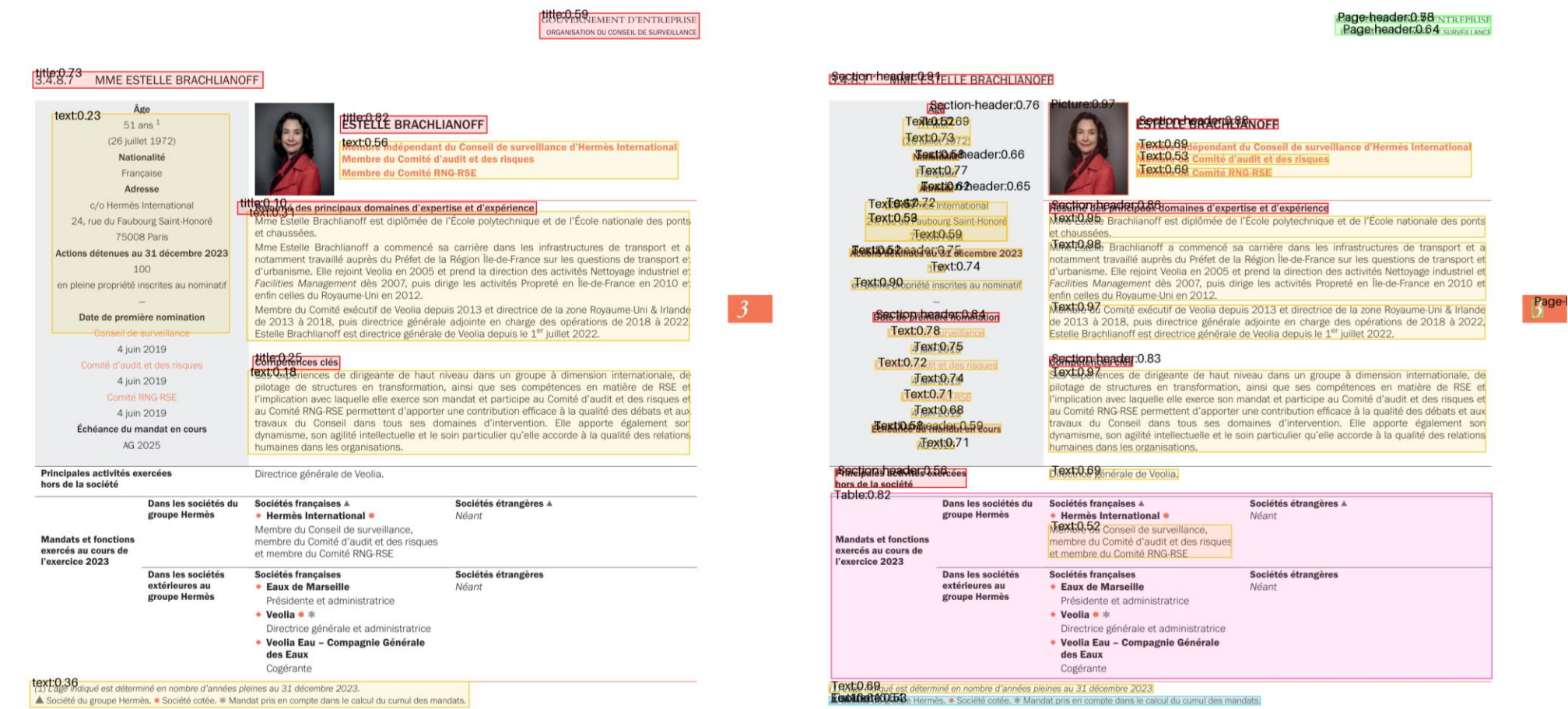


Docling eval

Simple-to-use and open evaluation suite



which segmentation?



of the moist paper. Maintain connection for 10 seconds while rocking the aluminum rod on the filter paper. Ensure that the light emitting diode (LED) remains lit (indicating good electrical contact and current flow) during this period. Disconnect the wiring assembly and set it aside. Remove the filter paper and examine it to determine that a light spot appears where the connection was made.

Deposit one drop of 1.0 percent solution of reagent grade dimethylglyoxime in ethyl alcohol on the filter paper (same side that was in contact with the test metal). A bright, distinctly pink spot will appear within seconds on the filter paper if the metal being tested is Inconel. A brown spot will appear if the test metal is stainless steel. Some stainless-steel alloys may leave a very light pink color. However, the shade and depth of color will be far less than would appear for Inconel. For flat surfaces, the test spot will be circular while for curved surfaces, such as the outside of a tube or pipe, the test spot may appear as a streak. (Refer to Figure 7-3 for sample test results.) This procedure should not be used in the heat-affected zone of weldments or on nickel coated surfaces.

Nonferrous Aircraft Metals

The term "nonferrous" refers to all metals that have elements other than iron as its base or principal constituent. This group includes metals, such as aluminum, titanium, copper, and magnesium, as well as alloyed metals, such as Monel and Babbitt.

Aluminum and Aluminum Alloys

Commercially pure aluminum is a white, lustrous metal, which stands second in the scale of malleability, sixth in ductility, and ranks high in its resistance to corrosion. Aluminum combined with various percentages of other metals forms alloys, which are used in aircraft construction. Aluminum alloys with principal alloying ingredients are manganese, chromium, or magnesium and silicon show little attack in corrosive environments. Alloys with which substantial percentages of copper are more susceptible to

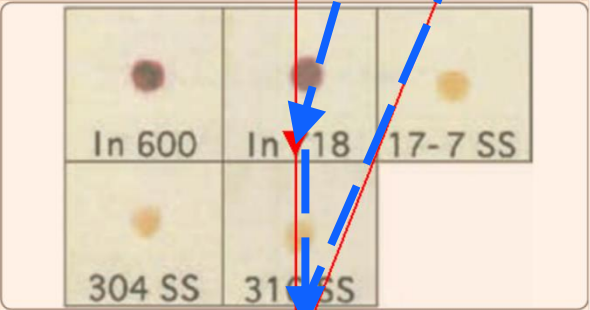


Figure 7-3. Electrochemical test results of Inconel (In) and stainless steel (SS) alloys.

corrosive action. The total percentage of alloying elements is seldom more than 6 or 7 percent in the wrought alloys.

Aluminum is one of the most widely used metals in modern aircraft construction. It is vital to the aviation industry because of its high strength-to-weight ratio and its comparative ease of fabrication. The outstanding characteristic of aluminum is its lightweight. Aluminum melts at the comparatively low temperature of 1,250 °F. It is nonmagnetic and is an excellent conductor.

Commercially pure aluminum has a tensile strength of about 13,000 psi, but rolling or other cold-working processes may approximately double its strength. By alloying with other metals, or by using heat-treating processes, the tensile strength may be raised to as high as 65,000 psi or to within the strength range of structural steel.

Aluminum alloys, although strong, are easily worked because they are malleable and ductile. They may be rolled into sheets as thin as 0.0017 inch or drawn into wire 0.004 inch in diameter. Most aluminum alloy sheet stock used in aircraft construction range from 0.016 to 0.096 inch in thickness; however, some of the larger aircraft use sheet stock that may be as thick as 0.356 inch.

The various types of aluminum may be divided into two general classes:

- Casting alloys (those suitable for casting in sand, permanent mold, or die castings)
- Wrought alloys (those which may be shaped by rolling, drawing, or forging).

Of these two, the wrought alloys are the most widely used in aircraft construction, being used for stringers, bulkheads, skin, rivets, and extruded sections.

Aluminum casting alloys are divided into two basic groups. In one, the physical properties of the alloys are determined by the alloying elements and cannot be changed after the metal is cast. In the other, the alloying elements make it possible to heat treat the casting to produce the desired physical properties.

A letter preceding the alloy number identifies the casting alloys. When a letter precedes a number, it indicates a slight variation in the composition of the original alloy. This variation in composition is simply to impart some desirable quality. For example, in casting alloy 214, the addition of zinc to improve its pouring qualities is indicated by the letter A in front of the number, thus creating the designation A214.

– Evaluation metrics tuned for documents

– Complete annotated datasets with all features
Layout structure, Reading order, Table structure, Text edits/additions, etc

– Easily identify tops and flops

– Compatibility and comparison with other evaluation and metrics

which reading order?

find model mistakes

Wrap-up

Recap

Docling:

- helps you **automate** your documents workflows with **ease**
- is **cost-effective**, **high-quality**, and respectful of data **privacy**
- drives strong **innovation** in model space and across the document intelligence space
- enables building powerful **AI applications**, from RAG & extraction to full-blown **agents**



Call for action

Help grow Docling and make it even better

1 Join the community

- Get started at docling.ai
- Star it on GitHub & spread the word
- Join Docling's LinkedIn, GitHub, Discord



docling.ai

2 Make a contribution

- Integrate into your favorite framework
- Add agentic examples & applications
- Support more input formats and connectors
- Become a C++ maintainer for docling-parse

